

4.d. HJB - Implementing Policy Iteration for Optimization and Pricing

Mean Variance Optimization (MVO) in Complete Markets Case

Portfolio optimization for risky assets like stock indices is often expressed as a Dynamic Programming Problem given it reduces to a Markov Decision Process (MDP). The optimal control policy (ϕ_t^* , the number of shares to hold in the portfolio) can be discretized using the Finite Difference Method (FDM) to simulate a stochastic control solution.

$$dS_t = (r + \xi\sigma_t)S_t dt + \sigma_t dW_t$$

where:

- W is the increment of a Wiener process,
- σ is volatility,
- ξ is the market price of risk,
- r is the risk-free rate.

Stochastic Linear-Quadratic Problem: The existence of a Dynamic Programming Principle (DPP) solution to the MDP is based on the concave properties of the utility/value functional U_t to be maximized. In the case of MVO, this is a quadratic function based on the first two moments of the random process X_t , ensuring convergence to an optimum value.

The portfolio process X_t and cash deposited in the pension account per unit of time π are governed by:

$$dX_t = \frac{\phi X_t}{S_t} dS_t + \pi dt + (1 - \phi)X_t r dt$$

Optimal Control ϕ_t^* Based Measure: The expectation for the HJB equation is taken with respect to the optimal wealth trajectory traced by the optimal control E^{ϕ^*} .

Risk Aversion: The risk aversion parameter γ is central to the intertemporal discounting of utility, and the coefficient of risk aversion λ serves as the Lagrangian operator used to transform the maximization problem:

$$\max_{\phi(t,x)} [E_{t=0}[X_T] - \lambda \text{Var}_{t=0}[X_T]]$$

subject to $\phi(t, X_t = x) \in Q$ (the set of all admissible strategies).

X_t is the wealth trajectory corresponding to the control $\phi(t, x)$.

The problem reduces to:

$$V(x, t) = \inf_{\phi \in Q} E [(X_T - 0.5\gamma)^2 | X_t = x] = \inf_{\phi \in Q} J(t, x, \phi)$$

The Resulting PDE:

$$V_t + \inf_{\phi \in Q} \left(\mu_x^\phi V_x + \frac{1}{2} (\sigma_x^\phi)^2 V_{xx} \right) = 0$$

where:

- $\mu_x^\phi = (r + \phi\xi\sigma)x + \pi$
- $\sigma_x^\phi = \phi\sigma x$

This PDE is localized to a finite interval $[0, x_{\max}]$ with a bound on wealth. The asymptotic boundary condition, in this case, is taken to be $V = 0$ when wealth w is large.

$$V_t + (rs + \pi)V_x = 0$$

Hamilton-Jacobi-Bellman (HJB) Solution:

The solution reduces to:

$$V(x, T - \tau) = \alpha(\tau)x^2 + \beta(\tau)x + \delta(\tau)$$

where:

- $\alpha(\tau) = e^{2r\tau}$
- $\beta(\tau) = -(\gamma + c)e^{r\tau} + ce^{2r\tau}$
- $\delta(\tau) = \left(\pi \frac{(\gamma+c)}{r} + \frac{\pi c}{2r}\right)(e^{r\tau} - 1) + \frac{\gamma^2}{4}$
- $\tau = T - t$

Indifference Pricing Temperature Call Option

Lastly, pricing a derivative or option based on its utility compared to the utility of a portfolio without the derivative is called *indifference pricing* in economics. This indifference price is implemented here in a simplified context. Control optimization on utility from a portfolio with a temperature derivative is equated to that of a utility from a portfolio without the derivative.

$$E_{(\phi^*, y)}(U[X_T]) = E_{(\phi^*, y)}(U[X_T + g(Y_T)])$$

where $g(Y_t, t)$ is the process representing the derivative's value over the time domain, depending on the Ornstein-Uhlenbeck (mean-reversion) temperature process Y_t . The indifference price of the derivative depends on:

$$\gamma \int_0^t g(y, t) dy$$

such that: